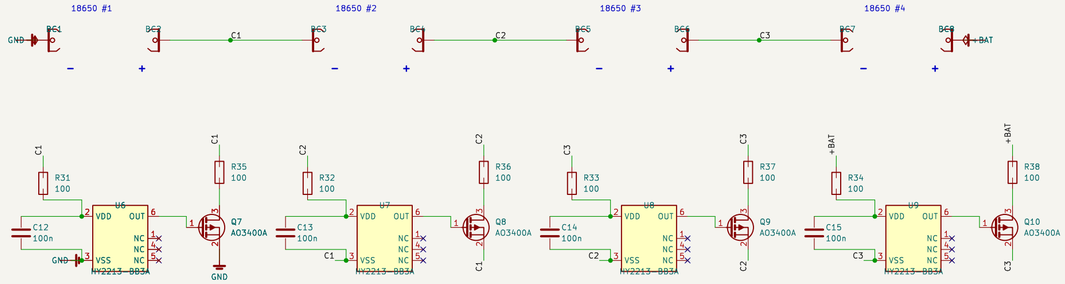
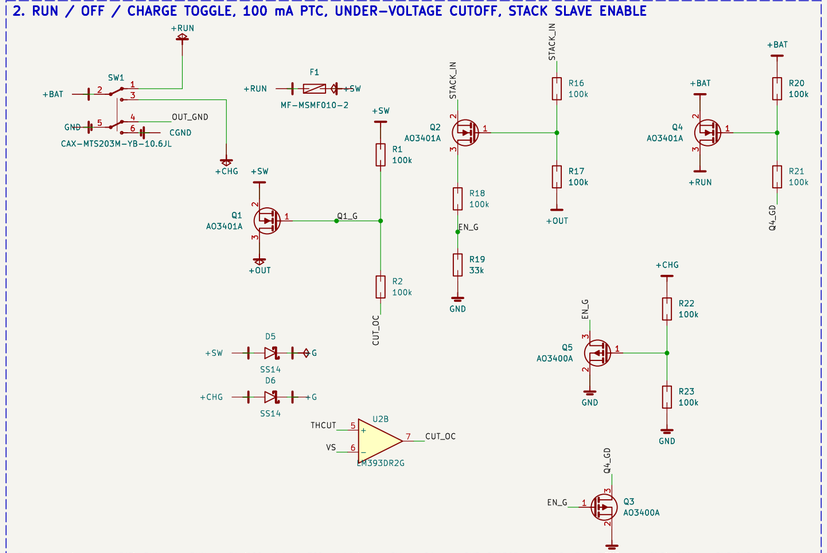
1. CELLS (4S 18650 in MY-18650-01 clips) AND PER-CELL BALANCERS (HY2213-BB3A, 42 mA above 4.20 V)



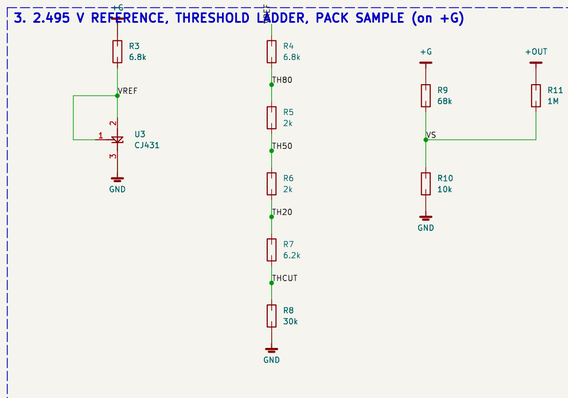
Each cell: two clips 48.0 mm apart (pin rows), contact plates 65 mm apart. Balancer per cell: HY2213 OUT drives an AO3400A that bleeds the cell through 100 R (42 mA, 0.18 W) above 4.200 V, off at 4.19 V; 2.5 uA standing. Together with the charger's 16.8 V this is the balance.

2. RUN / OFF / CHARGE TOGGLE, 100 mA PTC, UNDER-VOLTAGE CUTOFF, STACK SLAVE ENABLE



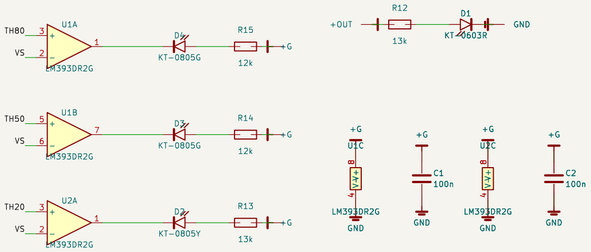
SW1 pole 1 (common +BAT): RUN -> +RUN -> F1 (100 mA hold, 300 mA trip) -> +SW -> Q1 -> +OUT: CHARGE -> +CHG. Pole 2 (common GND): RUN -> OUT_GND (J3.2): CHARGE -> CGND (USB-C ground, charger return). OFF opens both. Gauge rail +G = +SW or +CHG through D5 / D6. U2B pulls Q1's gate low (R1/R2: Vgs = -(+SW)/2) while VS > THCUT: +OUT off below 12.0 V, back on above 12.8 V (R11). Slave: with the upper board in RUN its +RUN arrives on J3.4 (STACK_IN), Q2 (gate on +OUT = upper GND) sources into Q3's gate, Q3 pulls Q4's gate down, Q4 feeds +RUN. Q5 blocks it in CHARGE.

3. 2.495 V REFERENCE, THRESHOLD LADDER, PACK SAMPLE (on +G)



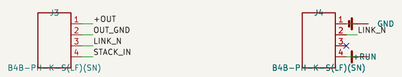
Ladder (107 uA): TH80 2.137 V, TH50 2.029 V, TH20 1.920 V, THCUT 1.595 V. VS = +G x 0.1357 with +OUT live (R9 || R11 against R10), x 0.1271 cut off: +G = +BAT - 0.25 V (SS14). Thresholds 16.0 / 15.2 / 14.4 V (pack) for D4 / D3 / D2, cutoff 12.0 V, re-enable 12.8 V. R3: >= 1.3 mA in U3 at 12 V.

4. GAUGE: THREE COMPARATORS, FOUR LEDs (about 1 mA each)



Open collectors sink the LED cathodes: D4 >= 16.0 V (80 %), D3 >= 15.2 V (50 %), D2 >= 14.4 V (20 %) from +G, so the bar also shows while charging; D1 (from +OUT) = output live. Bar reads the open-circuit voltage (the e-stim module draws ~10 mA). C1 / C2 at pin 8 of U1 / U2.

5. OUTPUT, STACK LINK, MOUNTING HOLES

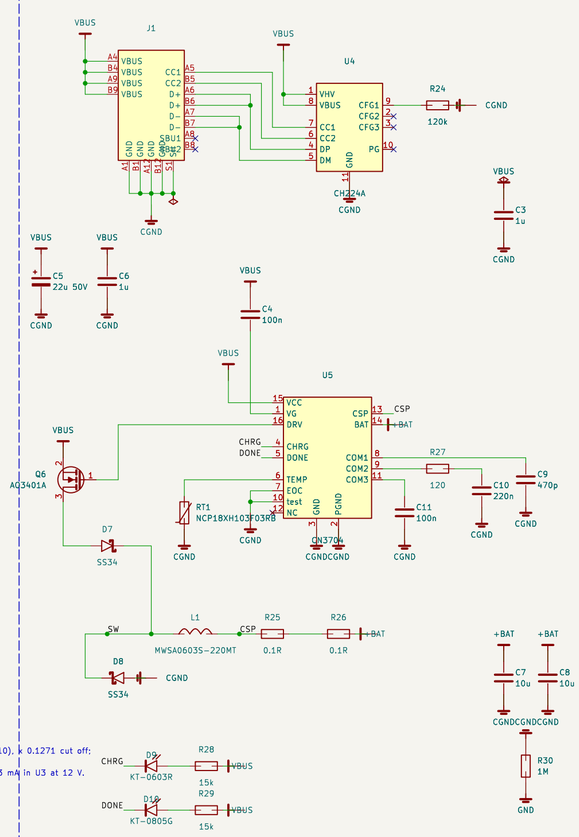


OUT: +V / 0V / -V / STACK_IN (to the module, or to the upper board's J4)

LINK IN from the lower board's J3 (lower +OUT -> GND here)

Two identical boards stacked on 25 mm M3 standoffs: lower J3 -> upper J4 with a straight 4-wire JST-PH cable: lower +OUT -> upper GND (= 0 V), lower OUT_GND -> LINK_N -> upper J3.3 (= -V), upper +RUN -> lower STACK_IN (slave enable). Upper J3 = +V / 0 V / -V (+RUN) to the e-stim module. Lower toggle: OFF normally (its output follows the upper toggle), CHARGE to charge; RUN only for a board on its own. Charging: either or both toggles in CHARGE, plug USB-C PD bricks in: a charging board has both output lines open, so grounds never loop.

6. USB-C PD 20 V INPUT (CH224A) AND 4S BUCK CHARGER (CN3704, 1 A), ALL ON CGND



J1 / U4: the CH224A asks a PD brick for 20 V (R24 120k on CFG1); below 20 V the CN3704 cannot reach 16.8 V and nothing charges. U5: buck at 300 kHz, Q6 switch, D7 blocks the pack from the input through Q6's body diode, D8 freewheels, L1 22 uH, R25 + R26 = 0.2 R -> 1 A constant current, 16.8 V +-1 % constant voltage, termination at 9 % (92 mA), trickle at 150 mA below 11.2 V, re-charge below 16.0 V, over-voltage cut at 18.1 V. RT1 10k NTC between the cells stops charging hot / cold (10k fixed resistor disables it). D9 CHRG (red), D10 DONE (green). CGND meets GND only in the CHARGE position (SW1 pole 2): R30 1M keeps it defined otherwise, so a brick plugged in while running does nothing. 20 uA from the pack idle.

Kemere lab
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Title: 4S 18650 battery pack for the LaserHAT e-stim module

Size: A2	Date: 2026-10-26	**Rev: B**
KiCad E.D.A. 9.0.8		Id: 1/1